

Product Requirements Document: California Coffee Market Expansion

1. Executive Summary

The goal of this project is to provide a data-driven framework for selecting the top 5 locations in California to establish a new coffee business. Additionally, the project includes a predictive model to determine the optimal price of a latte based on local demographic variables.

2. Problem Statement

Entering the saturated California coffee market requires more than "good coffee." Success depends on:

- Identifying neighborhoods with high disposable income but low "Third Wave" coffee saturation.
- Pricing products competitively yet profitably based on local purchasing power and demographic expectations.

3. Target Locations (Preliminary Top 5)

Based on 2026 market trends, the following regions are prioritized for analysis:

1. **San Francisco (SoMa/Mission):** High tech-worker density; premium pricing tolerance.
2. **Berkeley (Downtown/Uptown):** Academic population; high volume, ethical sourcing preference.
3. **Santa Cruz:** Coastal tourism mixed with remote-work professionals; high "craft" appreciation.
4. **Long Beach:** Rapidly gentrifying coastal area with high average coffee spending per capita.
5. **Oakland (Temescal/Rockridge):** Cultural hub for Gen Z and Millennials; high demand for functional/wellness coffee.

4. Feature Requirements

4.1. Location Identification Engine

- **Metric 1: Competition Density:** Number of independent vs. chain coffee shops within a 2-mile radius.
- **Metric 2: Foot Traffic Proxies:** Proximity to public transit hubs, universities, and co-working spaces.
- **Metric 3: Growth Velocity:** Year-over-year increase in local commercial rent prices as a proxy for neighborhood desirability.

4.2. Latte Price Predictive Model

The model will predict the price (P) of a standard 12oz latte using the following features:

- **Median Household Income (I):** Strong positive correlation with price ceiling.
- **Education Level (E):** Percentage of population with a Bachelor's degree or higher (proxy for specialty coffee interest).
- **Age Demographic (A):** Concentration of 20–40-year-olds (Gen Z/Millennials).
- **Local Rent (R):** Operational cost overhead adjustment.

Proposed Model Formula:

$$P_{latte} = \beta_0 + \beta_1(I) + \beta_2(E) + \beta_3(A) + \beta_4(R) + \epsilon$$

5. Data Sources

- **US Census Bureau / ACS:** For income, age, and education data.
- **Yelp/Google Places API:** For competitor mapping and existing price points.
- **Zillow/LoopNet:** For commercial real estate cost indexing.
- **Bureau of Labor Statistics (BLS):** For regional labor cost adjustments.

6. Success Metrics

- **Accuracy:** Predictive model within $\pm \$0.25$ of actual market leaders' pricing.
- **ROI Potential:** Identification of at least two "undervalued" neighborhoods where rent is low but demographics match high-value consumers.

7. Implementation Roadmap

1. **Phase 1 (Data Collection):** Scrape and aggregate demographic and competitor data.
2. **Phase 2 (Clustering):** Use K-Means clustering to identify "Coffee Hotspots."
3. **Phase 3 (Regression):** Train the pricing model on existing California coffee shop data.
4. **Phase 4 (Final Selection):** Rank locations based on a weighted score of competition, cost, and projected revenue.